Francis Padua

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Education

University of California, Irvine GPA: 3.97
Bachelor of Science in Computer Science *Expected Graduation: Jun. 2028*

Experience

Resmed Jun. 2026 – Sep. 2026
Platform Engineering Intern *San Diego, CA*

- Secured 60+ CI/CD workflows against shell injection vulnerabilities by building a detection library from 0 to 1
- Abstracted EKS deployment complexity by transpiling intent-based configs to Helm values via TypeScript and Zod
- Engineered a GitOps deployment layer with ArgoCD, cutting deployment latency from 5 minutes to 5 seconds

AntAlmanac Nov. 2025 – Present
Lead Software Engineer *Irvine, CA*

- Built UCI's open-source course scheduler (∼17K users), leading design and release direction for 12 developers
- Reduced search latency 30% by precomputing availability indexes, rendering live data via React and TypeScript
- Automated quarterly course-data releases with GitHub Actions workflow, taking 3–4 manual PRs per term to zero

Cyber@UCI, National Collegiate Cyber Defense Competition Nov. 2025 – Present
Software Developer, Competitor *Irvine, CA*

- Emulated complex attack paths by architecting intentionally vulnerable enterprise Linux topologies via Proxmox
- Enforced least-privilege network guardrails across 50+ critical Linux and WindowsOS services under active attack
- Engineered Ansible automations to lock down vulnerable services across networks, neutralizing active backdoors

California State University, Fullerton Jun. 2025 – Aug. 2025
Undergraduate Researcher *Fullerton, CA*

- Engineered a hybrid C++/Python pipeline to execute large-scale statistical arbitrage tests across 10K+ stock pairs
- Optimized computation by designing a C++ correlation filter to cull non-viable pairs, reducing the search space

Projects

Northstar | *Go, React, TypeScript, ConnectRPC, SQLite, GORM* Jan. 2026

- Built a real-time host operations dashboard by packing a React/Vite UI and SQLite database into single Go binary
- Redesigned credential handling to store zero passwords in the dashboard, mitigating secret leaks on compromise
- Eliminated stale telemetry for dashboard operators by resolving WebSocket race conditions via buffered channels

Hypernova | *Go, Model Context Protocol (MCP), ConnectRPC* Feb. 2026

- Engineered an MCP server in Go enabling AI agents to access critical service hardening and threat detection tools
- Exposed SSH command execution, playbook automation, log parsing, and Northstar APIs as typed tools for agents
- Designed credentials to be out of agents' context via index references, and gated mutating tools behind approval

Awards & Competitions

National Collegiate Cyber Defense Competition Apr. 2026

- **1st Place** in National Wild Card
- **8th Place** in Nationals

Western Region Collegiate Cyber Defense Competition Feb. 2026 – Mar. 2026

- **1st Place** in Qualifiers
- **2nd Place** in Regionals

Technical Skills

Languages: Go, C++, Python, TypeScript, JavaScript, SQL, Bash
Frameworks: React, Next.js, tRPC, ConnectRPC, Protobuf, Node.js, Vite, MUI, Tailwind CSS, Drizzle
Infrastructure: Kubernetes, Helm, Envoy Gateway, Kyverno, AWS, Ansible, Proxmox, GitHub Actions, Gravitee